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20 April 1984
SUPERSEDING
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MILITARY SPECIFICATION

EXPANSION JOINTS, PIPE, METALLIC BELLOWS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers piping system bellows expansion joints of metallic construction used to absorb dimensional movement caused by thermal expansion, contraction, and adjacent structural movements.

1.2 Classification. The expansion joints shall be of the following types and classes as specified (see 6.2), and shall be the size specified (see 6.2).

Type I - Corrugated bellows, unreinforced.
Type II - Corrugated bellows, reinforced.
Type III - Fabricated bellows.
Type IV - Internally-externally guided bellows.

Class 1 - Single bellows expansion joint.
Class 2 - Double bellows expansion joint.
Class 3 - Pressure balanced expansion joint.

Style 1 - Internally guided-externally guided.
Style 2 - Internally guided.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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MIL-P-116 - Preservation, Methods of.
MIL-M-7298 - Manuals, Technical: Commercial Equipment.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedure and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B16.5 - Steel Pipe Flanges and Flanged Fittings.
B31.1 - Code for Pressure Piping.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A240 - Heat Resisting Chromium and Chromium Nickel Stainless Steel Plate, Sheet and Strip for Fusion Welded Unfired Pressure Vessels.
B36 - Brass Plate, Sheet Strip and Rolled Bar.
B103 - Phosphor Bronze Plate, Sheet, Strip and Rolled Bar.
B127 - Nickel-Copper Alloy Plate, Sheet, and Strip.
B152 - Copper Sheet, Strip, Plate, and Rolled Bar.
B168 - Nickel-Chromium-Iron Alloy Plate, Sheet and Strip.
B194 - Copper-Beryllium Alloy Plate, Sheet, Strip, and Rolled Bar.
B409 - Nickel-Iron-Chromium Alloy Plate, Sheet and Strip.
B424 - Nickel-Iron-Chromium-Molybdenum-Copper Alloy Plate, Sheet and Strip.
B443 - Nickel-Chromium-Molybdenum-Columbium Alloy Plate, Sheet, and Strip.
E94 - Radiographic Testing.
E165 - Liquid Penetrant Inspection Method.
E709 - Magnetic Particle Examination.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Section VIII, Division 1 of ASME for Unfired Pressure Vessels and Code Case 1177-7.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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EXPANSION JOINT MANUFACTURERS ASSOCIATION, INC. (EJMA)

Standards of the Expansion Joint Manufacturers Association.

(Application for copies should be addressed to the Expansion Joint Manufacturers Association, 25 North Broadway, Tarrytown, NY 10591.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish within the time limit specified in the contract or order, a first article joint to determine conformance with this specification. The test joint shall consist of bellows, reinforcing or equalizing rings, if required, and appropriate end connections. Liners, covers, limit rods, tie rods, hinges, gimbals, and other similar devices need not be provided on the first article test joint, unless they are inherently required for the design and operation of the expansion joint assembly.

3.1.1 Determination of movement. In determining the qualifying rated contraction or extension, consideration shall be given to the equivalent axial movement due to lateral deflection and angular rotation. Equivalent axial movement shall be added to the actual specified values of axial movement and shall be computed in accordance with the EJMA Standards.

3.1.2 Universal, swing, and pressure-balanced expansion joints. Universal, swing, and pressure-balanced expansion joints consist of two or more bellows connected by pipe sections or fittings. Since it is not the intent of this specification to test these units as assemblies, only one prototype bellows is required to qualify any such combination of bellows.

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3 Standard commercial product. Each expansion joint of the same classification shall, as a minimum, be in accordance with the requirements of

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this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the joint being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Identical items. All joints of the same classification, furnished under any specific contract, shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable without prior written approval of the contracting officer.

3.5 Description. A bellows expansion joint shall consist of one or more bellows with suitable connections and accessories, as required, by the application. Terminology utilized in this specification is per the EJMA Standards.

3.5.1 Bellows. Material for the bellows shall consist of one or more of the following:

a. ASTM A240 - Types 304, 304L, 310, 316, 316L, 321, or 347.
b. ASTM B36 - 70/30 Brass (UNS C26000).
c. ASTM B103 - Grade A Phosphor Bronze (UNSC51100).
d. ASTM B127 - Class A Nickel Copper Alloy.
e. ASTM B152 - UNS C11000, C11300, C11400, C11600, C12000, C12200, C12300, C12500, or C14200.
f. ASTM B168 - Nickel-Chromium-Iron Alloy.
g. ASTM B194 - Beryllium Copper.
h. ASTM B409 - Nickel-Iron-Chromium Alloy.
i. ASTM B424 - Nickel-Iron-Chromium-Molybdenum-Copper Alloy.
j. ASTM B443 - Nickel-Columbiuim-Molybdenum Alloy.

Material not definitely specified shall be of the quality used for the purpose in commercial practice (see 6.2).

3.5.2 Types.

3.5.2.1 Type I. Type I expansion joints shall have bellows formed in accordance with 3.7.1.

3.5.2.2 Type II. Type II expansion joints shall meet the requirements of type I, and where required by 1.2, shall be provided with reinforcing rings or equalizing rings as described in 3.7.10.

3.5.2.3 Type III. Type III expansion joints shall have fabricated bellows of disk or diaphram construction.

3.5.2.4 Type IV. Type IV joints use an internally or externally guided bellows.

3.5.3 Styles.

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3.5.3.1 Style 1. Joint shall have guides located externally and internally to the chamber. The guides shall be integral with or attached to the body, or rigidly attached to the tube, and shall be concentric with each other, with the body, and with the tube. Limit stops shall be furnished in accordance with the manufacturer's standard practice.

3.5.3.2 Style 2. Joints shall have guides located internally to the chamber. The guides shall be integral with or attached to the body, or rigidly attached to the tube, and shall be concentric with each other, with the body, and with the tube. Limit stops shall be furnished in accordance with the manufacturer's standard practice.

3.5.4 Classes.

3.5.4.1 Class 1. Class 1 expansion joint shall have a single bellows.

3.5.4.2 Class 2. Class 2 expansion joint shall have two bellows separated by a section of pipe.

3.5.4.3 Class 3. Class 3 expansion joint shall be a pressure balanced joint in which the pressure thrust of the line bellows is offset by the effect of a balancing bellows.

3.6 Performance.

3.6.1 Movement. The expansion joint shall be capable of absorbing the normal operating movement, such as axial, lateral, angular, or a combination of these movements as specified (see 6.2 and 6.3).

3.6.1.1 Additional movement. When specified (see 6.2), the expansion joint shall be physically capable of absorbing an additional movement in an amount up to the rated movement of the joint (see 6.3 b.). This movement can be installation tolerances and/or upset conditions; the nature of this movement must be specified (see 6.2).

3.6.1.2 Cycle life expectancy. The capacity of the expansion joint to withstand the number of cycles specified shall be determined in accordance with the EJMA Standards, a history of successful operation of similar bellows, or by a qualification test, and shall be specified (see 6.2). The manufacturer shall certify compliance with these requirements.

3.6.2 Deflection force. The force required to deflect the expansion joint shall be equal to or less than the force specified (see 6.2). The connecting piping shall be properly guided and anchored in accordance with the EJMA Standards.

3.6.3 Expansion joint stability. The expansion joint shall be so designed as to preclude instability when the joint is subjected to the specified operating and/or test conditions. Instability shall be as defined in the EJMA Standards.

3.7 Construction. The expansion joints covered by this specification shall be designed and constructed in accordance with the generally accepted engineering principles set forth in ANSI B31.1, section VIII, Division 1, of the ASME Unfired Pressure Vessel Code, and the EJMA Standards.

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3.7.1 Corrugated bellows. Corrugated bellows for types I and II expansion joints shall be formed by one of the methods described in the EJMA Standards. Sizes up to and including 10 inches shall be formed from seamless tubing or from a tube having only one longitudinal welded seam. For sizes larger than 10 inches, more than one longitudinal welded seam is permissible. In no case will circumferential seams be accepted on formed corrugated bellows unless the requirements of a particular application require such construction and the design is approved by the procuring organization. Corrugations shall be symmetrical and of similar dimensions.

3.7.2 Fabricated bellows. Fabricated bellows for type III joints shall be constructed of single-formed diaphragms joined together with circumferential seams at the inner and outer perimeters of the diaphragms, or constructed of single formed corrugations joined together with circumferential seams of the inner perimeters of the corrugations.

3.7.3 Internally-externally guided bellows. The bellows in type IV expansion joints shall be designed and manufactured in accordance with the EJMA Standards. Sizes up to and including 10 inches shall be formed from seamless tubing or from a tube having only one longitudinal welded seam. For sizes larger than 10 inches, more than one longitudinal welded seam is permissible. In no case will circumferential seams be accepted unless the requirements of a particular application require such construction and the design is approved by the agency concerned. Convolutions shall be symmetrical and of similar dimensions.

3.7.4 End connections. The connecting ends of the joint shall be provided with pipe flanges or welding nipples as specified (see 6.2), Flanges shall conform to the diameter and drilling of ANSI B16.5. The size, material, and pressure class shall be as specified (see 6.2). Facings shall be as specified. Unless otherwise specified (see 6.2), the manufacturer may furnish "Van Stone" facings, made by turning or flaring the ends of the bellows over the flange face. The face of the Van Stone, however, is not a machine finish and it may not comply in every respect with the ANSI dimensions for a raised face. Welding nipples shall be of the size, material, and nominal wall thickness designation specified, and shall be beveled for welding.

3.7.5 Limit stops. When required for proper operation (see 6.2), limit stops shall be provided and arranged to limit the movement of the joint to that for which it is designed.

3.7.6 Guiding and alinement. Piping adjacent to the expansion joint shall be properly alined and guided so that the expansion joint is subjected to only those movements for which it is designed. Type II joints designed for the absorption of axial movement shall, when required, be provided with suitable alinement rods, guide sleeves, or covers to prevent distortion of the expansion joint by weight of the reinforcement hardware. Joints alined by internal guides shall have suitable means provided to prevent excessive increase or decrease in pressure between the internal guide and the bellows when the joint is subjected to movement. (Note: The use of internally or externally guided expansion joints does not eliminate the necessity of using adequate external pipe guides).

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3.7.7 Internal sleeves. Internal sleeves and material for these sleeves shall be specified for all expansion joints, regardless of the metal of the bellows, in the following cases (see 6.2):

a. When friction losses should be held to a minimum and smooth flow is desired.
b. When flow velocities exceed those specified in the EJMA Standards.
c. When there is danger of pitting or erosion, as in lines carrying abrasive solid particles, heavy sleeves are required.
d. During high temperature service with an uninsulated bellows, when it is necessary to decrease the temperature of the bellows and enable the metal to retain its higher mechanical properties.
e. When copper alloy bellows are used for high pressure drip, or superheated steam, hot water, or condensate, or when there is any possibility of flashing.

Internal sleeves should not be used when high viscosity fluids, such as tars, etc., are being transmitted, since these fluids may cause "packing-up", "coking", and "caking" which in turn may cause premature expansion joint failure. When the fluid is such that purging will effectively prevent "packing-up", internal sleeves may be used in conjunction with purging connections. Lightweight internal sleeves should not be considered as substitutes for internal guide sleeves.

3.7.8 Base. When specified (see 6.2), class 1 expansion joints shall be provided with a base. The base for class 1 joints shall be designed primarily as a support for the joint and when specified, shall also be designed for use as a main anchor. All expected forces and moments for anchor bases must be specified. Class 2 expansion joints shall be provided with a base suitable as an intermediate anchor.

3.7.9 Drain connection. When specified (see 6.2), a drain connection shall be provided on the center pipe section of class 2 expansion joints and shall be threaded, flanged, or beveled for welding.

3.7.10 Rings. Type II expansion joints shall be provided with reinforcing or equalizing rings. Reinforcing rings, which support the root of the corrugation under internal pressure, are solid or tubular in cross section. Equalizing rings, which support the root and sidewalls of the corrugation under internal pressure are usually "T" shaped in cross section. Equalizing rings, sometimes called control rings, have diameters larger than the outside diameter of the corrugations and limit the maximum compression of each corrugation.

3.7.11 Protective covers. When specified (see 6.2), sheet metal protective covers shall be provided which shall protect the bellows from external damage, prevent foreign material from falling between the corrugations of the bellows, and permit insulation of the expansion joint.

3.8 Bellows material.

3.8.1 Copper alloy bellows. Copper alloy bellows shall have their longitudinal seams joined by either welding or brazing operations. The bellows

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(except for those fabricated of diaphragms welded or brazed at inner and outer perimeters) shall be supplied in the stress relieved or annealed condition and shall be pickled to remove the heat treating scale. The maximum working pressure rating shall not exceed 25 pound-force per square inch gage (psig) for type I joints or 200 psig for type II joints.

3.8.2 Corrosion-resistant steel bellows. Corrosion-resistant steel bellows shall have their longitudinal seams joined by welding. The bellows may be stress relieved or annealed at the discretion of the manufacturer. When some form of heat treating is employed, however, the bellows shall be pickled to remove the resultant scale. If iron or carbon steel forming tools are used with corrosion-resistant steel bellows, the bellows shall be passivated after forming.

3.8.3 Other bellows materials. If materials other than those mentioned previously are used, the bellows shall be constructed in accordance with the manufacturer's standard practice.

3.9 Drawings. A dimensional sketch or drawing in sufficient detail to describe the equipment which the contractor proposes to furnish shall be submitted to the agency concerned with the bid proposal. Such a sketch or drawing shall be accompanied by a copy of the manufacturer's specification and rating data providing sufficient detail to signify compliance with the requirements of this specification.

3.10 Treatment and painting. All external surfaces subject to corrosion, except flange faces and threaded connections, shall be treated and painted in accordance with the manufacturer's best standard practice.

3.11 Technical publications. When specified (see 6.2), a manual conforming to the requirements of MIL-M-7298 and applicable options shall be furnished with each expansion joint. The manual shall include data necessary for the proper installation, operation, and maintenance of the expansion joint.

3.12 Workmanship. The quality of workmanship shall be such as to produce equipment that is in accordance with the requirements of this specification and insure proper functioning of all parts of the unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete joint when a first article is required (see 3.1 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the joint meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining joints to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall be performed on sample expansion joints selected in accordance with 4.3 and the packaging inspection of 4.6.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete expansion joint. All joints of the same type, class, material, design, and movement per corrugation offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level of 2.5 percent defective.

4.3.2 Sampling for tests. A random sample of joints shall be selected in accordance with table I from each lot and tested as specified in 4.5, with acceptance based on the following requirements in accordance with the procedures specified in MIL-STD-105.

TABLE I. Sampling for testing.

Number in lot	Number in sample	Acceptance number	Rejection number
5 and under	all	0	1
6 to 100	5	0	1

4.4 Examination. Each joint selected shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects that prevents a joint from functioning with maximum efficiency shall constitute cause for rejection.

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4.5 Tests. The following tests shall be performed. Failure of any joint to pass any specified test shall be cause for rejection.

4.5.1 Hydrostatic test. Before cycling, each joint shall be subjected to a hydrostatic pressure equal to the maximum test pressure specified for any of the joints. This test shall be performed first with the joint compressed to its qualifying rated contraction and then with the joint extended to its rated extension. When no extension is specified, this second test shall be made with the joint in its neutral position.

4.5.2 Cyclic endurance test. When specified (see 6.2), a cyclic endurance test shall be performed on joints selected with the number of complete cycles as specified (see 6.2). The test shall be performed under pressure at ambient temperature and the joint shall be cycled in axial movement only. During the test, the pressure in the joint shall be adjusted to simulate, as closely as possible, the maximum design conditions of the joint to be qualified. Thus, the maximum design pressure specified for the group shall be adjusted in accordance with paragraph UG-99, section VIII of the ASME Code, based on the ratio of the stress values for the bellows material. The pressure may vary from zero to this value during each cycle, providing the pressure reaches its maximum when the joint is compressed to its qualifying rated contraction. Upon completion on the required number of cycles, each test joint shall be pressure-tested as before. If the test joint fails to pass this cyclic endurance test, it shall be rejected and a new joint of improved design shall be fabricated and tested. Any change in the test joint must be reflected in the final production units.

4.5.3 Pressure and vacuum testing. Roll formed bellows shall be pressure or vacuum tested. Bellows formed by hydraulic force shall not require testing provided that they have not been subjected to a welding or brazing operation. Hydraulically formed bellows which develop leaks during forming shall be rejected. Fabricated bellows shall be tested when specified (see 6.2).

4.5.3.1 Joints rated for pressure service. When the design pressure is equal to or larger than 15 psig the joint shall be subjected to a hydrostatic test pressure equal to 1-1/2 times the design pressure. When the design pressure is less than 15 psig an air and soap leak test shall be performed in lieu of the hydrostatic pressure test.

4.5.3.2 Joints rated for vacuum service. Vacuum service assemblies shall be tested pneumatically to the rated vacuum of the expansion joint; or shall be tested pneumatically or hydrostatically at a pressure above the atmosphere equivalent to 1-1/2 times the rated vacuum.

4.5.4 Shock test. When specified (see 6.2), a shock test shall be required for joints delivered for shipboard use.

4.5.5 Other tests. A die penetrant test conforming to ASTM E165, a radiographic examination conforming to ASTM E94, or a magnetic particle test conforming ASTM E269 may be specified in addition to pressure or vacuum testing (see 6.2).

4.5.6 First article test report. The manufacturer shall submit to the contracting officer a complete report covering each first article test performed. Each report shall include, but is not limited to, the following data:

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a. The drawing number of the expansion joint.
b. Mill test reports of materials used in the joint.
c. Hydrostatic test pressure.
d. The qualifying rated contraction and extension, or both values.
e. Report of the number of complete cycles obtained.
f. Report of pressure test, or other leakage test, performed after completion of the cyclic endurance test.
g. The adjusted internal pressure during the cyclic endurance test.

4.6 Packaging inspection. The preservation, packing, and marking of the unit shall be inspected to determine conformance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-V-3 with the level of preservation and the level of packing as specified (see 6.2). In addition, the following requirements shall apply:

5.1.1 Instructions. Instructions for each joint shall be preserved method IC-3 in accordance with MIL-P-116.

5.1.2 Preparation for shipment. Expansion joints shall be prepared for shipment in a manner that maintains the proper end-to-end dimensions required for installation of the joint. When no temperature is specified, the expansion joint shall be prepared for shipment with end-to-end dimensions in accordance with the manufacturer's standard practice. Each joint shall be accompanied by a complete set of instructions for compression or extension to the correct end-to-end dimension of the joint at the existing temperature during installation. These instructions shall be attached to the joint and shall be fully protected against moisture.

6. NOTES

6.1 Intended use. Bellows expansion joints covered by this specification are intended for steam, water, oil, or gas piping services which require the absorption of movement resulting from the thermal expansion of piping or other factors.

6.1.1 Types of joints. The types of bellows expansion joints listed under 1.2 are basic constructions. Arrangements of this basic type vary with the individual requirements of each application. The most common arrangement is one which absorbs axial movement. Joints of this type are sometimes suitable for the absorption of small amounts of lateral deflection. Movement other than axial motion can be absorbed by expansion joints of special arrangement as indicated by the following examples:

a. Hinged expansion joint - An expansion joint designed to permit angular rotation in one plane only by the use of a pair of pins through hinge plates attached to the expansion joint ends.
b. Gimbal expansion joint - An expansion joint designed to permit angular rotation in any plane by use of two pairs of hinges affixed to a common floating gimbal ring.

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c. Universal expansion joint - An expansion joint containing two bellows joined by a common connector for the purpose of absorbing any combination of the three basic movements, i.e., axial movement, lateral deflection, and angular rotation.
d. Swing expansion joint - An expansion joint designed to absorb lateral deflection and angular rotation in one plane only by the use of a pair of swing bars.
e. Pressure-balanced expansion joint - An expansion joint designed to absorb axial movement and lateral deflection while absorbing the pressure thrust by means of tie devices interconnecting the flow bellows with an opposed bellows also subjected to line pressure.

6.1.2 Pressure. Copper alloy expansion joints should be restricted to the pressure conditions specified in 3.7.1. Type I expansion joints (except copper (see 3.7.1)) are commonly limited to low-pressure services, with the upper limit being between 30 and 100 psig working pressure depending on size and the specific design. Type II and III expansion joints (except copper alloy (see 3.7.1)) are suitable for medium and higher pressures. All types are suitable for vacuum service.

6.1.3 Sizes. Expansion joints most often all custom engineered and fabricated for specific applications, because there is an infinite number of sizes that might be required for use in piping systems which include both standard and nonstandard sizes, specific sizes are not cited in the classification section 1.2 of this specification.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, style, and size required (see 1.2).
c. First article if required for inspection and approval (see 3.1, 4.2.1, and 6.5).
d. Bellows material (see 3.5.1).
e. Performance (see 3.6.1).

(1) Movement required for normal operation (see 3.6.1).
(2) Additional movement required (see 3.6.1.1).
(3) Number of cycles of movement for which joint designed (see 3.6.1.2).

f. Deflection force, if other than specified (see 3.6.2).
g. End connections (see 3.7.4).
h. If "Van Stone" facings are required and size of material and pressure class (see 3.7.4).
i. Whether limit stops are to be provided (see 3.7.5).
j. Internal sleeves and material, if required (see 3.7.7).
k. Base, if required, for class 1 expansion joints; or if not required for class 2 expansion joints, and if base is suitable as an end anchor (see 3.7.8).
l. Drain connection, type, and preparation, if required (see 3.7.9).
m. Protective cover, if required (see 3.7.11).
n. Technical publications and detailed requirements, if different (see 3.11 and 6.2.2).
o. When a cyclic endurance test is required for first article and production units and the number of complete cycles required (see 4.5.2).

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p. Test pressure if other than specified (see 4.5.3).
q. Shock test if required (see 4.5.4).
r. Other tests if required (see 4.5.5).
s. Level of preservation and level of packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 Performance. The following notes pertain to 3.6 and subparagraph thereto:

a. Packless expansion joint design is dependent upon several variable conditions under which the joint is intended to operate. It is therefore imperative that all applicable conditions under which the joint is to operate be specified in acquisition documents because the joints must, to insure adequate flexibility, be designed with a lower factor of safety compared to usual piping equipment.
b. Motion. The type and the degree of motion which the joint is required to absorb (see 3.6.1) shall be specified separately from the requirement for extra traverse motion (see 3.6.1.1). It is suggested that expansion joints shall be specified to be suitable for 25 percent more movement than the calculated expansion of the piping. The type and amount of motion can be specified directly, but a dimensioned and labeled sketch or drawing of the piping layout is generally preferred by the manufacturer to permit full study of the design requirements. A drawing or sketch will permit the manufacturer to recommend anchorage, guidance, alinement requirements, and other recommendations for the specific application.
c. Cyclic endurance (see 3.6.1.2). The life of a packless expansion joint depends upon several variables of which the following are a few samples: thickness of bellows material, size of the bellows, number of corrugations, amount of motion, frequency of motion, temperature and pressure conditions. Therefore, no life expectancy can be generally specified for all types of joints. The life requirement is best specified by the number of complete cycles of movement which is anticipated for the period the joint should last. This would be determined by the number of times the service would be shut down over the required period, or by the number of times the temperature would vary between certain limits during the period. Vibrational movement should be specified by amplitude and frequency as flexural stresses in the bellows under vibration should be within the proportional limit of the material used. Cyclic endurance proof tests are expensive since the joint under test is not suitable for use after the test. Such tests merely indicate suitability of design and should only be specified where critical conditions warrant the extra time and expense incurred.

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6.4 Test pressure. Most expansion joint manufacturers usually test their joints at 1-1/2 times rated pressure (see 6.2 n.). Joints can be manufactured for test at pressures higher than 1-1/2 times the rated pressure, but to do so will usually force the design to be based upon the test pressure rather than on the operating pressure. Such practice would in most cases require an increase in thickness of the bellows material and a corresponding increase in the number of corrugations to realize the cyclic endurance life. The characteristics of a particular application should govern the decision to alter the test pressure requirements specified in 6.2.

6.5 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one expansion joint. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing Activity:

Navy - YD

(Project 4730-0425)

Review Activities:

Navy - SH
Air Force - 82

User Activities:

Army - CE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side) 4730-0425

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-17813E	EXPANSION JOINTS, PIPE, METALLIC BELLOWS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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